

PREDICTING AIRBNB PRICES

# Sevilla, Andalucía

# Price Prediction Project

Machine Learning · SoSe 2026

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**Tutorial 1 · Group 7**

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# Objective & Framework

## Goal

Build a predictive model that estimates Airbnb listing prices, combining several data modalities and modeling approaches.

## City & Data

**Sevilla, Andalucía, Spain**

Downloaded: listings, calendar, reviews (Inside Airbnb)

### Modalities Used

**Tabular**

listing and calendar

**Text**

reviews

**Spatial**

listing coordinates (latitude / longitude)

# Data Sources at a Glance

| Modality | Source File  | Key Fields / Features                                                    |
|----------|--------------|--------------------------------------------------------------------------|
| Tabular  | listings.csv | price, room_type, accommodates, bathrooms, bedrooms, reviews, host stats |
| Tabular  | calendar.csv | booked days, weekday/weekend split, seasonal availability                |
| Text     | reviews.csv  | multilingual sentiment score & label per listing                         |
| Spatial  | listings.csv | latitude, longitude, neighbourhood_cleansed                              |

**Final dataset sizes:** listings\_clean.csv → 7,426 listings × 30 columns | combined\_data.csv → 7,426 listings × 44 columns

Price range: 29€ – 1,145€ · mean ≈ 152€; median = 125€

# Data Cleaning & Preprocessing

- 1 Cleaned price: removed currency symbols, converted to numeric
- 2 Trimmed outliers — dropped top & bottom 1% of prices
- 3 Selected 25 relevant columns for modeling
- 4 Parsed `host_response_rate` ("% " removed) → numeric, median-imputed
- 5 Imputed missing `host_is_superhost` via most-frequent category
- 6 Dropped rows missing `bathrooms` / `bedrooms` / `host_total_listings_count`
- 7 Median-imputed remaining review-score NaNs
- 8 Filled missing text fields (name, description, amenities) with empty strings

Downloaded data:

Listings: 8215 rows, 79 columns

Calendar: 2998475 rows, 7 columns

Reviews: 685106 rows, 6 columns



After preprocessing and feature extraction:  
7,426 accommodations and 44 features

# Feature Engineering

## Listings

- Binary-encoded  
instant\_bookable,  
host\_is\_superhost
- One-Hot encoding for  
room\_type
- Label encoding for  
property\_type &  
neighbourhood\_cleansed

## Calendar

- Total booked days per listing
- Booked weekday vs. weekend  
days
- Booked days per season
- Booked days relative to  
availability

## Reviews

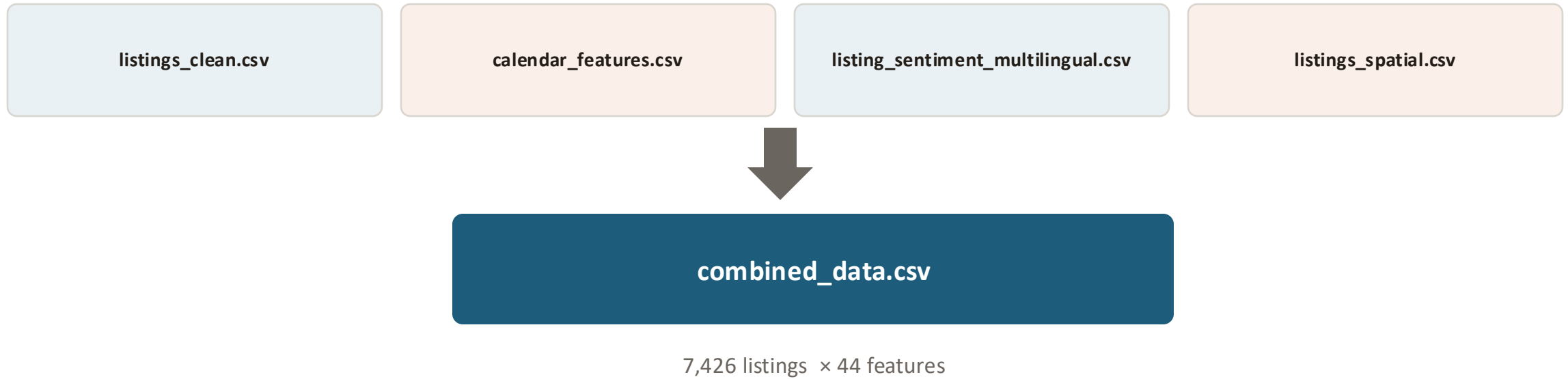
- Multilingual sentiment scoring  
per review (negative, neutral,  
positive)
- Aggregated  
avg\_sentiment\_score per listing (if  
95% sure that positive then score  
is +95%)

## Spatial

- Latitude / longitude carried as  
numeric features
- Combined with neighbourhood  
encoding

# Data Integration

Four processed sources merged on listing ID:



Purpose: enables direct comparison between tabular-only models and multimodal models trained on the merged, feature-rich dataset.

# Modeling Approaches

1

## Random Forest

Baseline tree ensemble,  
trained on listings-only and  
combined data

*n\_estimators = 100*

2

## XGBoost

Gradient-boosted trees;  
primary predictor  
minimizing residual error

*learning\_rate = 0.1,  
n\_estimators = 200*

3

## CatBoost (tuned)

Gradient-boosted trees with  
native categorical handling,  
hyperparameter tuned

*learning\_rate = 0.20, depth =  
10*

4

## k-Nearest Neighbors

Distance-based baseline  
with grid-search tuning

*tuned: n\_neighbors, weights,  
p*

5

## Residual Correction

Second-stage model trained  
on residuals to correct  
systematic bias

*stacked on top of best base  
predictor*

**Evaluation: 80 / 20 train-test split · metrics = MAE (Mean Absolute Error) and RMSE (Root Mean Squared Error)**

# Price Correction

1

## CatBoost

Use tuned CatBoost model

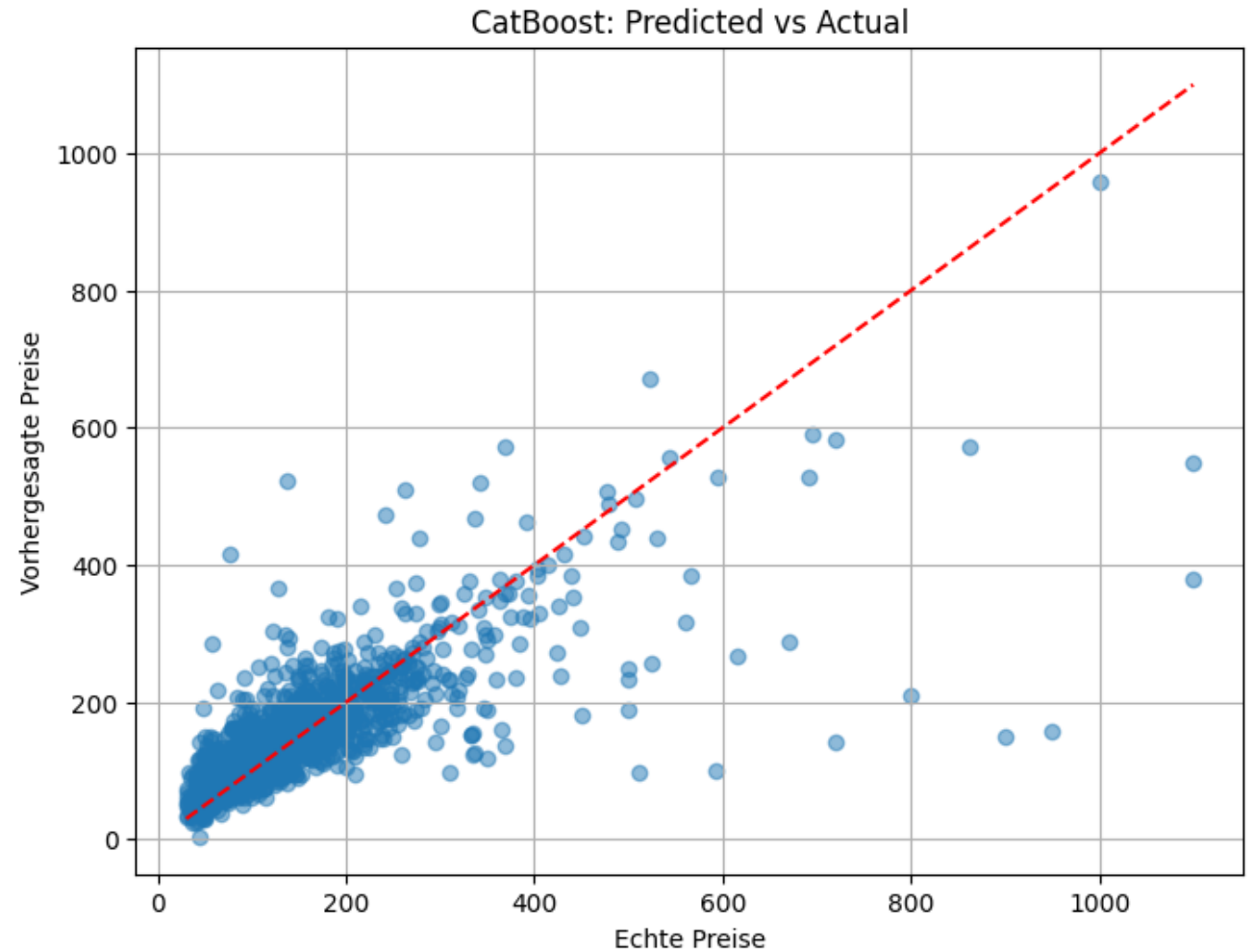
*learning\_rate = 0.20, depth = 10*

2

## Linear correction

Linear regression with predicted price and actual price.

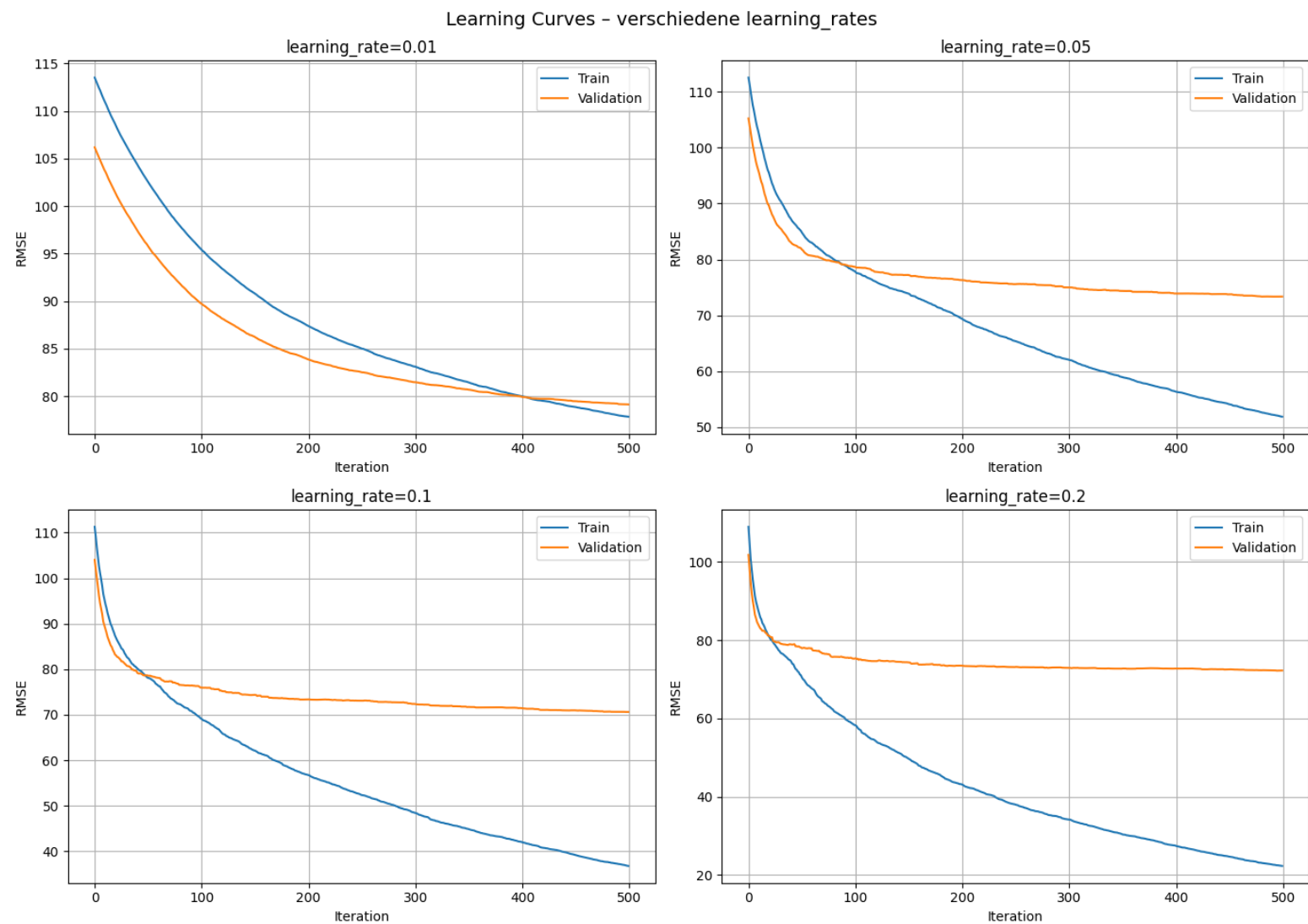
Other models:  
Isotonic  
Catboost (X\_train, residuals)



**Doesn't work, CatBoost already works pretty well**

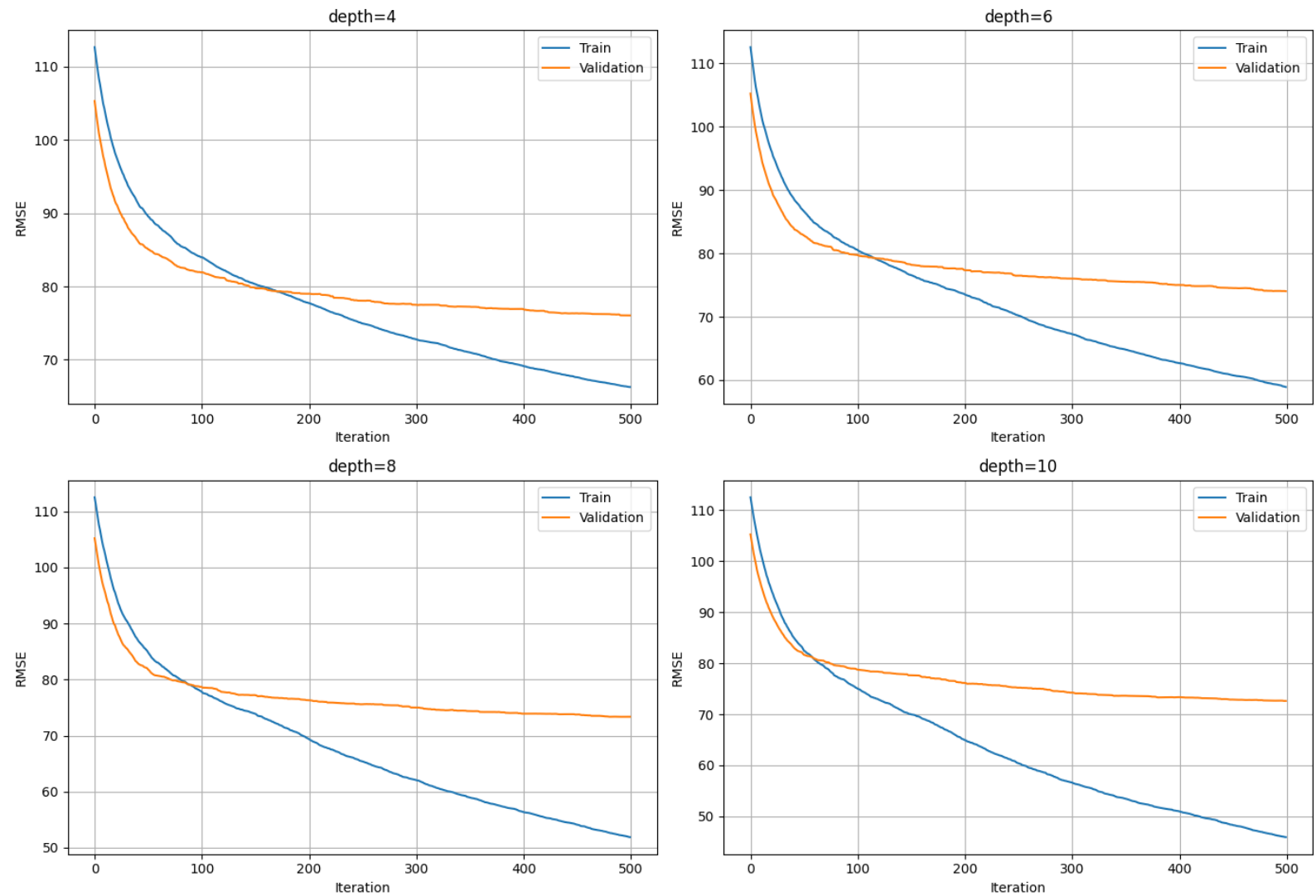


# CatBoost tuning

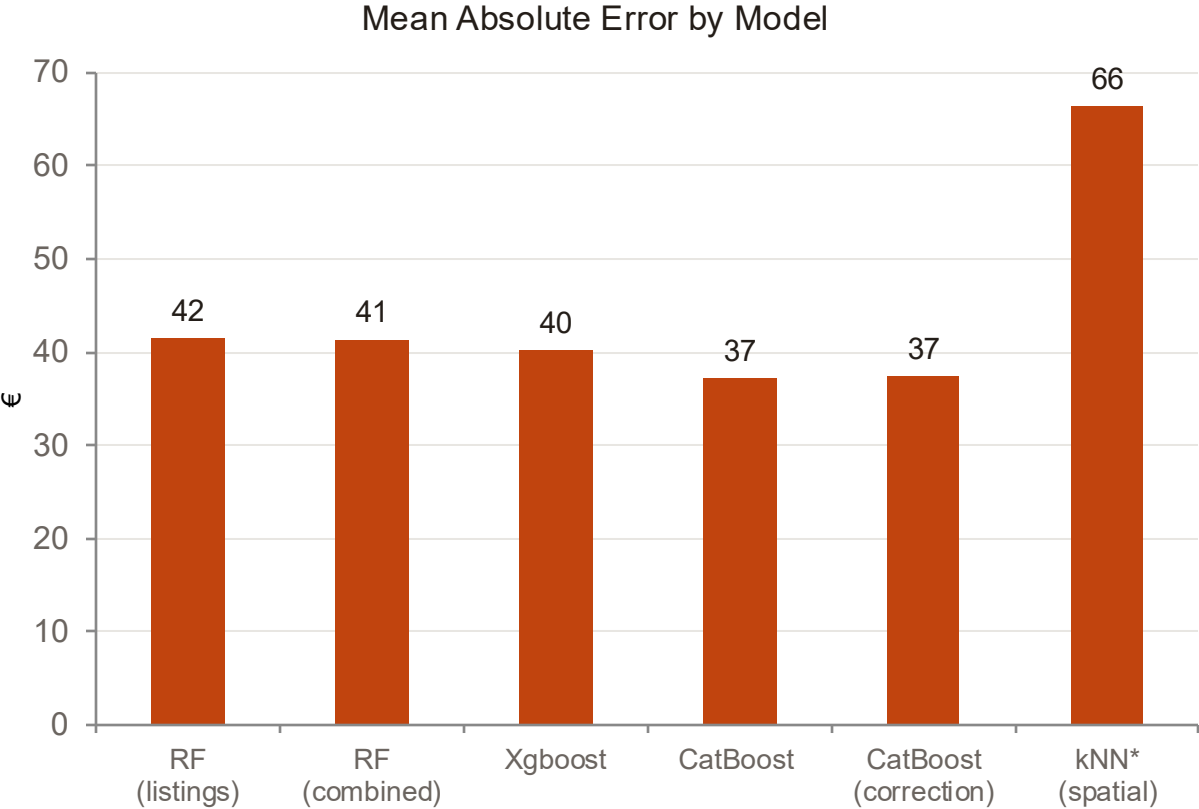


# CatBoost tuning

Learning Curves - verschiedene depths



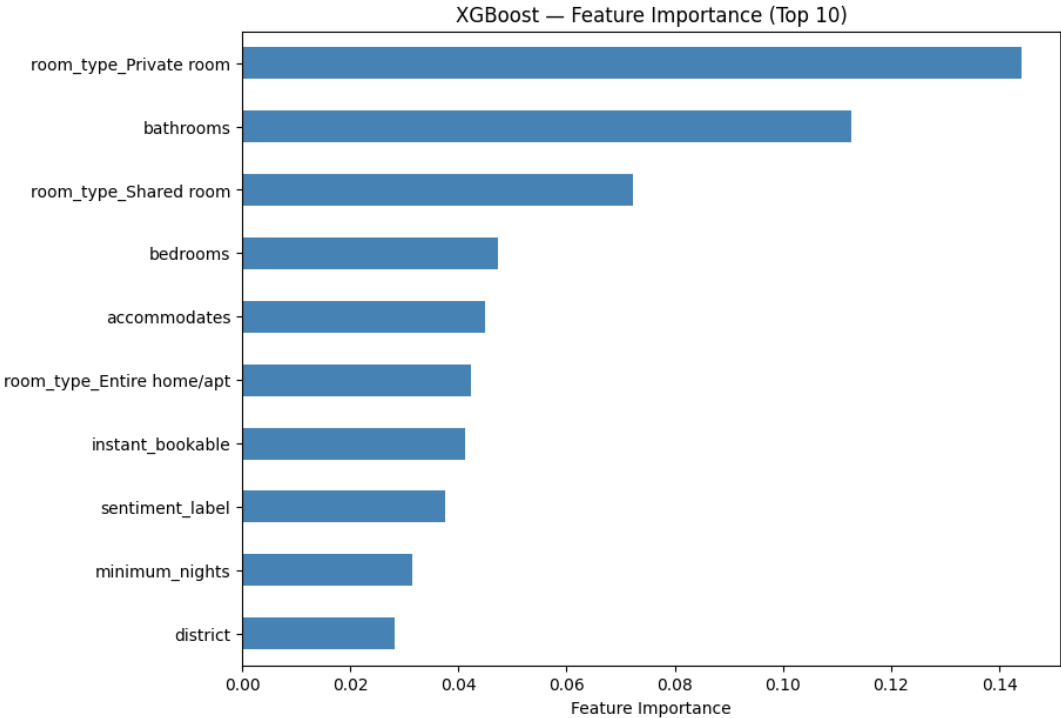
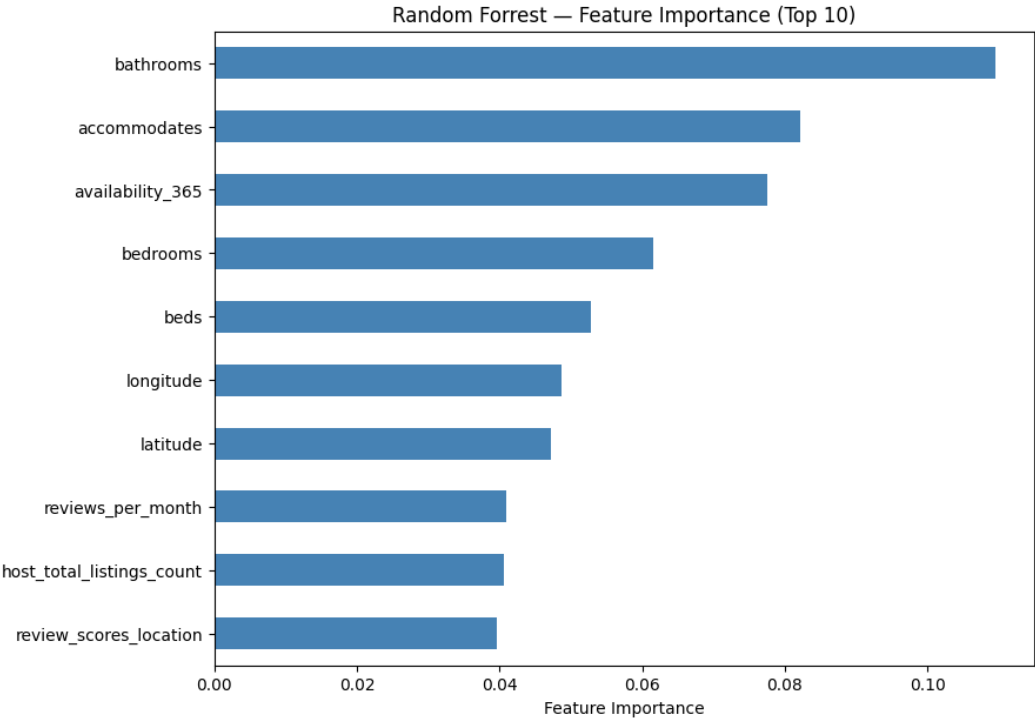
# Model Comparison



| Model                 | MAE (€) | RMSE (€) |
|-----------------------|---------|----------|
| Random Forest         | 41.50   | 77.00    |
| Random Forest (tuned) | 41.38   | 75.32    |
| Xgboost               | 40.29   | 73.86    |
| CatBoost (tuned)      | 37.28   | 70.6     |
| CatBoost (correction) | 37.42   | 70.62    |
| kNN (tuned)*          | 66.37*  | —        |

*\* kNN value is cross-validated MAE, not held-out test MAE — not directly comparable. Its untuned baseline scored  $R^2 = -0.06$  (worse than predicting the mean price).*

# Feature Importance



# Discussion & Next Steps

## Key Findings

- CatBoost on combined data performs best: MAE 37.42€, RMSE 70.6€
- Merging calendar, sentiment & spatial data cut RF's RMSE by ~23% (99.60€ → 76.74€)
- bathrooms, accommodates and location consistently drive price
- avg\_sentiment\_score enters the top-10 features only once text data is added
- kNN underperforms tree-based models on this high-dimensional mixed data

## Limitations

- Image modality not yet used (photos referenced only by URL)
- High-price outliers (>1,000€) add noise despite 1% trimming
- kNN evaluated via CV-MAE, not directly comparable to test MAE

## Next Steps

- Add image embeddings (e.g. CNN features from listing photos)
- Try hybrid model: text embeddings + tree-based regressor
- Broaden hyperparameter search for CatBoost & kNN